Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 04

Subject: BSC102/AP | **Date: 26/10/2026**
Max Marks: 25 | **Deadline: 09/11/2026**

CO4: To describe Maxwell's equations and their significance.

Q1. Gradient, Divergence and Curl

(a) Scalar and Vector Fields; The Del Operator

- **Scalar Field:** A scalar field is a region in space where each point (x, y, z) is assigned a single scalar physical quantity (characterized solely by magnitude). Examples include temperature distribution $T(x, y, z)$ and electric potential $V(x, y, z)$.
- **Vector Field:** A vector field is a region in space where each point (x, y, z) is assigned a vector physical quantity (having both magnitude and direction). Examples include the electric field $\mathbf{E}(x, y, z)$ and fluid velocity field $\mathbf{v}(x, y, z)$.
- **Del Operator (∇):** In Cartesian coordinates, the del (nabla) operator is a vector differential operator defined as:

$$\nabla = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

(b) Calculation of $\nabla\phi$

Given scalar field:

$$\phi(x, y, z) = 3x^2y - y^3z^2$$

The gradient is given by:

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

Computing the partial derivatives:

$$\frac{\partial\phi}{\partial x} = \frac{\partial}{\partial x}(3x^2y - y^3z^2) = 6xy$$
$$\frac{\partial\phi}{\partial y} = \frac{\partial}{\partial y}(3x^2y - y^3z^2) = 3x^2 - 3y^2z^2$$
$$\frac{\partial\phi}{\partial z} = \frac{\partial}{\partial z}(3x^2y - y^3z^2) = -2y^3z$$

Evaluating at the point $(-1, -2, 1)$:

$$\left.\frac{\partial \phi}{\partial x}\right|_{(-1,-2,1)} = 6(-1)(-2) = 12$$
$$\left.\frac{\partial \phi}{\partial y}\right|_{(-1,-2,1)} = 3(-1)^2 - 3(-2)^2(1)^2 = 3(1) - 3(4)(1) = 3 - 12 = -9$$
$$\left.\frac{\partial \phi}{\partial z}\right|_{(-1,-2,1)} = -2(-2)^3(1) = -2(-8)(1) = 16$$

Therefore:

$$\nabla\phi(-1, -2, 1) = 12\hat{i} - 9\hat{j} + 16\hat{k}$$

(c) Divergence of a Vector Field

Definition: The divergence of a vector field $\mathbf{F}$ measures the net outward flux of the field per unit volume from an infinitesimal region around a given point:

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Given vector field:

$$\mathbf{F} = xz\hat{i} + y^2z^3\hat{j} - xyz\hat{k}$$

Evaluating the components:

$$\frac{\partial F_x}{\partial x} = \frac{\partial}{\partial x}(xz) = z$$
$$\frac{\partial F_y}{\partial y} = \frac{\partial}{\partial y}(y^2z^3) = 2yz^3$$
$$\frac{\partial F_z}{\partial z} = \frac{\partial}{\partial z}(-xyz) = -xy$$

Thus:

$$\nabla \cdot \mathbf{F} = z + 2yz^3 - xy$$

Evaluating at the point $(3, -1, 2)$:

$$\nabla \cdot \mathbf{F}|_{(3,-1,2)} = (2) + 2(-1)(2)^3 - (3)(-1) = 2 - 16 + 3 = -11$$

Physical Interpretation: Since $\nabla \cdot \mathbf{F} = -11 < 0$, the vector field converges at the point $(3, -1, 2)$. This point acts as a **sink**, meaning more field lines enter an infinitesimal volume enclosing this point than leave it.

(d) Curl of a Vector and Identity $\nabla \cdot (\nabla \times \mathbf{A}) = 0$

Explanation and Significance:

- The curl of a vector field $\mathbf{A}$, denoted by $\nabla \times \mathbf{A}$, represents the infinitesimal rotation or circulation density of the field per unit area around a given point.

- **Significance:** It indicates whether a vector field possesses rotational characteristics or vortices. If $\nabla \times \mathbf{A} = 0$, the field is called *irrotational* (conservative), which guarantees that $\mathbf{A}$ can be expressed as the gradient of a scalar potential ($\mathbf{A} = -\nabla V$).

Proof that $\nabla \cdot (\nabla \times \mathbf{A}) = 0$: Let $\mathbf{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$. The curl of $\mathbf{A}$ is:

$$\nabla \times \mathbf{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{k}$$

Taking the divergence of $\nabla \times \mathbf{A}$:

$$\nabla \cdot (\nabla \times \mathbf{A}) = \frac{\partial}{\partial x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

Expanding the partial derivatives:

$$\nabla \cdot (\nabla \times \mathbf{A}) = \frac{\partial^2 A_z}{\partial x \partial y} - \frac{\partial^2 A_y}{\partial x \partial z} + \frac{\partial^2 A_x}{\partial y \partial z} - \frac{\partial^2 A_z}{\partial y \partial x} + \frac{\partial^2 A_y}{\partial z \partial x} - \frac{\partial^2 A_x}{\partial z \partial y}$$

By Clairaut's theorem, assuming the components of $\mathbf{A}$ possess continuous second partial derivatives, mixed partials commute:

$$\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}, \quad \frac{\partial^2}{\partial x \partial z} = \frac{\partial^2}{\partial z \partial x}, \quad \frac{\partial^2}{\partial y \partial z} = \frac{\partial^2}{\partial z \partial y}$$

All pairs cancel out identically:

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

Q2. Maxwell's Equations

(a) Gauss's Laws for Static Electric and Magnetic Fields

1. Gauss's Law for Static Electric Fields

- **Statement:** The total outward electric flux through any closed Gaussian surface is proportional to the net charge enclosed within that surface.
- **Integral Form:**

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon_0} = \frac{1}{\varepsilon_0} \iiint_V \rho \, dV$$

- **Differential Form:** Using the Divergence Theorem ($\oint_S \mathbf{E} \cdot d\mathbf{A} = \iiint_V (\nabla \cdot \mathbf{E})\, dV$):

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

2. Gauss's Law for Static Magnetic Fields

- **Statement:** The net magnetic flux through any closed surface is identically zero, implying that isolated magnetic poles (monopoles) do not exist.
- **Integral Form:**

$$\oint_S \mathbf{B} \cdot d\mathbf{A} = 0$$

- **Differential Form:** Applying the Divergence Theorem:

$$\nabla \cdot \mathbf{B} = 0$$

(b) Maxwell's Equation for Time-Varying Magnetic Fields (Faraday's Law)

Statement: The induced electromotive force (EMF) around a closed loop is equal to the negative time rate of change of magnetic flux enclosed by that loop.
Derivation: In integral form, Faraday's law is given by:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt}$$

Since magnetic flux is $\Phi_B = \iint_S \mathbf{B} \cdot d\mathbf{A}$:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{A} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{A}$$

Applying Stokes' Theorem to the closed loop integral of $\mathbf{E}$:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{A}$$

Equating both expressions:

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{A} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{A} \implies \iint_S \left(\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{A} = 0$$

Because this holds for any arbitrary surface S, the integrand must vanish:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

(c) Ampere's Circuital Law for a Static Magnetic Field

Statement: The line integral of magnetic field induction $\mathbf{B}$ along any closed loop is equal to μ_0 times the total steady current passing through the surface enclosed by that loop.

- **Integral Form:**

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} = \mu_0 \iint_S \mathbf{J} \cdot d\mathbf{A}$$

  where $\mathbf{J}$ is the volume current density.

- **Differential Form:** Applying Stokes' Theorem to the left-hand side:

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{B}) \cdot d\mathbf{A}$$

Equating this with the right-hand side:

$$\iint_S (\nabla \times \mathbf{B}) \cdot d\mathbf{A} = \iint_S \mu_0 \mathbf{J} \cdot d\mathbf{A} \implies \iint_S (\nabla \times \mathbf{B} - \mu_0 \mathbf{J}) \cdot d\mathbf{A} = 0$$

Since the open surface S is arbitrary:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$